3.25 Valuation Rings and Places

Definition 3.25.1 (Valuation Ring)
*A subring $A \subset K$ of a field K is a **valuation ring for K** if for every $0 \neq x \in K$ either $x \in A$ or $x^{-1} \in A$ (or both). Such a ring is an integral domain and K is necessarily a quotient field for A.*

*An integral domain A is a **valuation ring** if it is a valuation ring for its quotient field.*

Proposition 3.25.2 (Properties of valuation rings)
Let A be a valuation ring and K its quotient field then the following properties hold

a) A is local ring
b) $x \in A \setminus A^\star \iff x \in \mathfrak{m} \iff x^{-1} \notin A$
c) A is integrally closed in K

Proof. We prove each in turn

a) By (3.20.3) we need to show that $\mathfrak{m} := A \setminus A^\star$ is an additive subgroup of A. Given $x, y \in \mathfrak{m}$, without loss of generality we may assume that x, y are non-zero, and $\frac{x}{y} \in A$. Then $x + y = y\left(1 + \frac{x}{y}\right)$. If $(x + y) \in A^\star$ then $y \in A^\star$ a contradiction. Therefore $(x + y) \in \mathfrak{m}$ as required.
b) Note $x^{-1} \notin A \iff x \in A \setminus A^\star \iff x \in \mathfrak{m}$.
c) Suppose $0 \neq x \in K$ is integral over A. If $x \in A$ we are done. If $x^{-1} \in A$ then by (3.24.13) $x \in A[x^{-1}] \subseteq A$ as required.

■

Lemma 3.25.3 (Restriction of a Valuation Ring to a Subfield)
Let A be a valuation ring of a field M and $L \subset M$ a subfield. Then $A \cap L$ is a valuation ring of L.

Proof. For $0 \neq x \in L \subset M$, either $x \in A$ or $x^{-1} \in A$; as $x, x^{-1} \in L$ too, the same alternative holds for $A \cap L$. ■

Proposition 3.25.4 (Local Homomorphism)
Let $\phi : A \to B$ be a homomorphism of local rings. Then the following are equivalent

a) $\phi(\mathfrak{m}_A) \subset \mathfrak{m}_B$
b) $\mathfrak{m}_A = \phi^{-1}(\mathfrak{m}_B)$
c) $\phi(\mathfrak{m}_A)B \subsetneq B$

*We say that in this case ϕ is a **local homomorphism**. When ϕ is simply inclusion then we say B **dominates** A.*

Proof. a) $\implies$ b) $\phi^{-1}(\mathfrak{m}_B)$ is a proper ideal containing $\mathfrak{m}_A$ and therefore equal by maximality

b) $\implies$ a) This is obvious

a) $\implies$ c) $\phi(\mathfrak{m}_A)B \subset \mathfrak{m}_B B = \mathfrak{m}_B \subsetneq B$

c) $\implies$ a) By (3.3.15) $\phi(\mathfrak{m}_A)B \subset \mathfrak{m}_B$ and trivially $\phi(\mathfrak{m}_A) \subset \phi(\mathfrak{m}_A)B$. ■

Corollary 3.25.5
Let $(A, \mathfrak{m})$ be a local ring, K a field and $\phi : A \to K$ a homomorphism. Then the following are equivalent

a) ϕ is a local ring homomorphism (i.e. $\mathfrak{m} \subset \ker(\phi)$)
b) $\ker(\phi) = \mathfrak{m}$
c) ϕ factors through $\kappa(\mathfrak{m})$.

Lemma 3.25.6 (Restriction to a Valuation Ring)
Let A be a valuation ring of a field F, $L \supseteq F$ a field and $B \subseteq L$ a subring containing A, with a proper ideal $\mathfrak{n} \subsetneq B$ such that $\mathfrak{m}_A \subseteq \mathfrak{n}$ (equivalently $\mathfrak{m}_A B \neq B$). Then $B \cap F = A$ and $\mathfrak{n} \cap F = \mathfrak{m}_A$. In particular this applies to any local ring $B \subseteq L$ dominating A (3.25.4), with $\mathfrak{n} = \mathfrak{m}_B$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Clearly $A \subseteq B \cap F$. Let $x \in F \setminus A$; then $x \neq 0$ and $(x^{-1})^{-1} = x \notin A$, so $x^{-1} \in \mathfrak{m}_A \subseteq \mathfrak{n}$ by (3.25.2). If $x \in B$ then $1 = xx^{-1} \in \mathfrak{n}$, which is absurd; so $x \notin B$ and $B \cap F = A$. Finally $\mathfrak{n} \cap F \subseteq B \cap F = A$ is a proper ideal of A, hence contained in $\mathfrak{m}_A$; the reverse inclusion is $\mathfrak{m}_A \subseteq \mathfrak{n}$. ■

Lemma 3.25.7 (A Directed Union of Valuation Rings is a Valuation Ring)
Let $(K_i)_{i \in I}$ be a directed system of fields with union $K := \bigcup_i K_i$, and for each i let $R_i \subset K_i$ be a valuation ring, such that R_l dominates R_i (3.25.4) whenever $K_i \subseteq K_l$. Then $R := \bigcup_i R_i$ is a valuation ring of K, with $\mathfrak{m}_R = \bigcup_i \mathfrak{m}_{R_i}$.

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Proof. R is a subring of K, being a directed union of subrings. For $0 \neq x \in K$, $x \in K_i$ for some i (directed union of fields); as R_i is a valuation ring of K_i, $x \in R_i$ or $x^{-1} \in R_i$, hence $x \in R$ or $x^{-1} \in R$: R is a valuation ring of K.

$\mathfrak{m}_R \subseteq \bigcup_i \mathfrak{m}_{R_i}$: if $x \in \mathfrak{m}_R$, then $x \in R_i$ for some i (as $R = \bigcup_i R_i$); as $x^{-1} \notin R \supseteq R_i$, in particular $x^{-1} \notin R_i$, so $x \in \mathfrak{m}_{R_i}$ (3.25.2) b).

$\mathfrak{m}_R \supseteq \bigcup_i \mathfrak{m}_{R_i}$: let $x \in \mathfrak{m}_{R_i}$, and suppose for contradiction $x^{-1} \in R_l$ for some l. By directedness choose m with $i, l \leq m$, so $x \in R_i \subseteq R_m$ and $x^{-1} \in R_l \subseteq R_m$, that is $x \in R_m^\star$. But R_m dominates R_i, so $\mathfrak{m}_{R_i} \subseteq \mathfrak{m}_{R_m}$, and $x \in \mathfrak{m}_{R_i}$ gives $x \in \mathfrak{m}_{R_m}$, contradicting $x \in R_m^\star$. So $x^{-1} \notin R_l$ for every l, that is $x^{-1} \notin R$, giving $x \in \mathfrak{m}_R$. ∎

Definition 3.25.8 (Place (Zariski-Samuel 1960 / Lang 1972))
*Let K be a field. A **place** of K consists of a valuation ring $(A, \mathfrak{m}_A)$ for K and a local homomorphism to a field F*

$$\phi : A \to F$$

(such that $\ker(\phi) = \mathfrak{m}_A$)

Furthermore if $x \in K \setminus A$ then we may write $\phi(x) = \infty$. Note that the second part of the previous Proposition then may be reinterpreted as saying

$$\phi(x) = \infty \iff \phi(x^{-1}) = 0 \quad \forall x \in K$$

which motivates the alternative definition below.

*We say it is a **semi-place** of K if $(A, \mathfrak{m}_A)$ is simply a local ring.*

Lemma 3.25.9
Let $A \subset K$ be a subring of a field and $\mathfrak{a} \lhd A$ a proper ideal. Then at least one of $\mathfrak{a}A[x]$ or $\mathfrak{a}A[x^{-1}]$ is a proper ideal.

Proof. Suppose neither are proper then we can write

$$1 = \sum_{j=0}^{n} a_j x^j$$

$$1 = \sum_{j=0}^{m} b_j x^{-j}$$

for $a_j, b_j \in \mathfrak{a}$. Choose n, m to be minimal and assume wlog that $m \leq n$. Observe that $a_0 \neq 1 \implies m > 0$. Multiply the second equation by $x^n a_n$ to find

$$x^n a_n (1 - b_0) = a_n b_1 x^{n-1} + \ldots a_n b_m x^{n-m}$$

and multiply the first by $(1 - b_0)$ to find

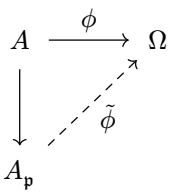
$$(1 - b_0) = a_0(1 - b_0) + \ldots + a_n(1 - b_0)x^n$$

consequently cancelling the x^n term and we obtain a relation of smaller degree a contradiction. ∎

We prove the first extension theorem

Proposition 3.25.10 (Extension to localization)
Let A be a ring and $\phi : A \to \Omega$ a homomorphism into a field. Let $\mathfrak{p} := \ker(\phi)$. Then

- *$\mathfrak{p}$ is prime*
- *There is a unique extension $\tilde{\phi}$ making the diagram commute*



Furthermore $\ker(\tilde{\phi}) = \mathfrak{p}A_\mathfrak{p}$ i.e. $\tilde{\phi}$ is local.

Proof. Clearly $\mathfrak{p}$ is prime because $\phi(A)$ is an integral domain. We may extend ϕ to the ring $A_\mathfrak{p}$ in the obvious way. The extension has kernel $\mathfrak{p}A_\mathfrak{p}$, the unique maximal ideal of $A_\mathfrak{p}$. ∎

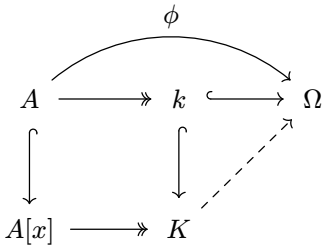
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Proposition 3.25.11 (Places as maximal extensions)
Let A be a subring of a field K and $\phi : A \to \Omega$ a homomorphism into an algebraically closed field. Then

- *For all $x \in K$, ϕ may be extended to at least one of $A[x]$ and $A[x^{-1}]$.*
- *There exists a maximal extension $\tilde{\phi} : B \to \Omega$. For any such maximal extension B is a valuation ring and $\tilde{\phi}$ is local (i.e. $\ker(\tilde{\phi}) = \mathfrak{m}_B$). Furthermore $\mathfrak{m}_B \cap A = \ker(\phi)$.*

Proof. We prove each in turn.

- By (3.25.10) we may assume without loss of generality that A is a local ring with unique maximal ideal $\mathfrak{m} = \ker(\phi)$. By (3.25.9) we may also suppose that $\mathfrak{m}A[x]$ is proper. Then it's contained in a maximal ideal $\mathfrak{B} \lhd A[x]$. Furthermore $\mathfrak{m} = \mathfrak{B} \cap A$ by maximality of $\mathfrak{m}$. Let $k = A/\mathfrak{m}$ and $K = A[x]/\mathfrak{B}$, then there is a commutative diagram



  Then $K = k[\overline{x}]$ is a field and by (...) $\overline{x}$ is algebraic over k. Therefore K/k is algebraic and, because Ω is algebraically closed, by (3.19.61) there is an extension to K, which gives the required extension to $A[x]$.
- It's easy to show that the poset of extensions to subrings of K ordered by consistency is chain complete. Therefore by Zorn's Lemma there is a maximal extension $\tilde{\phi} : B \to \Omega$. By the previous part for any $x \in K$ any such maximal element B must satisfy either $B[x] = B$ or $B[x^{-1}] = B$, i.e. B is a valuation ring for K. Consider $\mathfrak{B} := \ker(\tilde{\phi})$ a prime ideal contained in $\mathfrak{m}_B$. Then by (3.25.10) may extend $\tilde{\phi}$ to $B_{\mathfrak{B}}$ and so by maximality $B = B_{\mathfrak{B}}$. Finally (3.20.6) shows that B is a local ring with maximal ideal $\mathfrak{B} = \mathfrak{m}_B$. Clearly $\ker(\tilde{\phi}) \cap A = \ker(\phi)$, so the final statement follows easily.

■

Corollary 3.25.12
Let $A \subset K$ be a subring of a field and $\mathfrak{a} \lhd A$ a proper ideal. Then there exists a valuation ring $(B, \mathfrak{m}_B)$ such that $A \subset B$ and $\mathfrak{a} \subset \mathfrak{m}_B \cap A$. In particular if $\mathfrak{a} = \mathfrak{m}_A$ is maximal then $\mathfrak{m}_A = \mathfrak{m}_B \cap A$.

Proof. By (3.3.37) there is a maximal ideal $\mathfrak{m}_A \lhd A$ containing $\mathfrak{a}$. Let $k = A/\mathfrak{m}_A$ and $\Omega = \overline{k}$. Then the canonical homomorphism $\phi : A \to \Omega$ has kernel $\mathfrak{m}_A$. It has an extension to a valuation ring $(B, \mathfrak{m}_B)$ by (3.25.11), such that $\mathfrak{m}_B \cap A = \ker(\phi) = \mathfrak{m}_A$. ■

Proposition 3.25.13 (Alternative Characterisation of Valuation Rings)
Let K be a field and $A \subset K$ a subring. Then the following are equivalent

a) A is a valuation ring for K
b) A is a local ring maximal under the relation "B dominates A"
c) $K = \mathrm{Frac}(A)$ and the principal ideals of A are totally ordered
d) $K = \mathrm{Frac}(A)$ and the ideals of A are totally ordered

Proof. Let $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$ denote the relation B dominates A.

a) $\implies$ b) Suppose that $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$ with B local and A a valuation ring. We claim that $A = B$; suppose for a contradiction that $x \in B \setminus A$. Then $x^{-1} \in A \implies x^{-1} \in B \implies x \in B^\star$. Therefore $A \subset B \setminus B^\star = \mathfrak{m}_B$. In particular $1 \in \mathfrak{m}_B$ which is a contradiction as $\mathfrak{m}_B$ is proper.

b) $\implies$ a) Let $(A, \mathfrak{m}_A)$ be a maximal local ring. By (3.25.12) there exists a valuation ring $(B, \mathfrak{m}_B)$ such that $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$. By maximality $A = B$ and A is a valuation ring.

a) $\implies$ c) Evidently $K = \mathrm{Frac}(A)$. Suppose $(a) \not\subset (b)$, then $a \notin (b) \implies b^{-1}a \notin A \implies a^{-1}b \in A \implies b \in (a) \implies (b) \subset (a)$.

c) $\implies$ d) Suppose $\mathfrak{a} \not\subset \mathfrak{b}$ then there exists $a \in \mathfrak{a} \setminus \mathfrak{b}$. In particular for all $b \in \mathfrak{b}$ we have $a \notin (b)$, whence by assumption $b \in (a)$. Therefore we conclude $\mathfrak{b} \subset (a) \subset \mathfrak{a}$.

d) $\implies$ b) As A is a non-zero ring it has a maximal ideal $\mathfrak{m}_A$, which by assumption must be the unique maximal ideal. So $(A, \mathfrak{m}_A)$ is a local ring. Suppose that $(A, \mathfrak{m}_A) \preceq (B, \mathfrak{m}_B)$ and given $x \in B$ we have by assumption $x = a_1 a_2^{-1}$ for

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some $a_1, a_2 \in A$. If $a_1 \in (a_2)$ then $x \in A$. Otherwise $a_2 \in (a_1)$ whence $x^{-1} \in A \implies x^{-1} \in B^\star \overset{(3.25.2)}{\implies} x^{-1} \notin \mathfrak{m}_B \overset{(3.25.4)}{\implies} x^{-1} \notin \mathfrak{m}_A \implies x^{-1} \in A^\star \implies x \in A$. Consequently $A = B$ and b) holds. ∎

Corollary 3.25.14
Let $A \subset K$ be a subring of a field then the integral closure of A in K (denoted $\overline{A}$) satisfies

$$\overline{A} = \bigcap_{A \subset V} V$$

where the intersection is taken over all valuation rings V of K containing A.

Alternatively the integral elements over A are precisely the elements which are finite at all places of K, which are finite over A.

Proof. First if $x \in \overline{A}$ then by (3.24.13) we have $x \in A[x^{-1}] \subseteq V[x^{-1}]$. If $x \notin V$ then by hypothesis $x^{-1} \in V$, whence $x \in V$ a contradiction. Therefore $x \in V$ as required.

Conversely suppose $x \notin \overline{A}$, then $x \notin A[x^{-1}]$. That is to say (x^{-1}) is a proper ideal in $A[x^{-1}]$. Therefore by (3.25.12) there is a valuation ring $(V, \mathfrak{m}_V)$ such that $x^{-1} \in \mathfrak{m}_V$ which implies $x \notin V$ by (3.25.2). ∎

Definition 3.25.15 (Valuation)
*Let K be a field and Γ an abelian ordered group. A function $v : K^\star \to \Gamma$ is said to be a **valuation** if it satisfies the following properties*

a) $v(ab) = v(a) + v(b)$ for all $a, b \in K^\star$
b) $v(a+b) \geq \min(v(a), v(b))$ with the convention that $v(0) = \infty$

Note $v(1) = 0$. We say that two valuations v, v' are equivalent if there is an order-preserving isomorphism $\phi : v(K^\star) \to v'(K^\star)$ such that $\phi \circ v = v'$. Clearly this is an equivalence relation.

If $k \subset K$ and $v(k) = 0$ then we say v is a valuation for K/k.

Lemma 3.25.16 (Strict Ultrametric Inequality)
Let $v : K^\star \to \Gamma$ be a valuation (3.25.15), extended to K by $v(0) := \infty$. For $x, y \in K$ with $v(x) \neq v(y)$,

$$v(x+y) = \min(v(x), v(y))$$

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

If $x = 0$ or $y = 0$ this is immediate ($v(0) = \infty$ dominates), so assume $x, y \in K^\star$, and without loss of generality $v(x) < v(y)$. By property 2 of (3.25.15), $v(x+y) \geq \min(v(x), v(y)) = v(x)$. For the reverse inequality, apply property 2 again to $x = (x+y) + (-y)$:

$$v(x) \geq \min(v(x+y), v(-y))$$

As $v(-1) + v(-1) = v((-1)(-1)) = v(1) = 0$, and Γ is totally ordered (so $v(-1) > 0$, resp. $v(-1) < 0$, would force $v(-1) + v(-1) > 0$, resp. < 0), we must have $v(-1) = 0$; so $v(-y) = v(-1) + v(y) = v(y)$. As $v(y) > v(x)$, the displayed inequality $v(x) \geq \min(v(x+y), v(-y))$ forces $v(x) \geq v(x+y)$ (else the minimum would be $v(-y) = v(y) > v(x)$, a contradiction). Combined with the first inequality, $v(x+y) = v(x) = \min(v(x), v(y))$. ∎

Proposition 3.25.17 (Correspondence between Valuations Rings and Valuations)
Let K be a field with $k \subset K$. Then there is a bijection

$$\begin{aligned} \{k \subset A \subset K \mid A \text{ is a valuation ring}\} &\longleftrightarrow \{v : K^\star \to \Gamma \mid v \text{ is a valuation over } k\}/\sim \\ A &\to \quad v_A : K^\star \to K^\star/A^\star \\ v^{-1}(\Gamma_{\geq 0}) &\leftarrow \quad v : K^\star \to \Gamma \end{aligned}$$

Under this map v_A is just the quotient map and $\overline{x} \leq \overline{y} \iff yx^{-1} \in A^\star$. Furthermore

a) $v_A(\overline{x}) \geq 0 \iff x \in A$.
b) $v_A(\overline{x}) = 0 \iff x \in A^\star$
c) $v_A(\overline{x}) > 0 \iff x \in \mathfrak{m}_A$

Proof. Given a valuation ring we may define the abelian group $\Gamma := K^\star/A^\star$ under multiplication with $\overline{x} \leq \overline{y} \iff yx^{-1} \in A$. This is easily verified to be a total ordering. Then $v_A : K^\star \to K^\star/A^\star$ is simply the quotient map and we clearly $v_A(\overline{x}) \geq 0 \iff x \in A^\star$. To demonstrate the additive condition we may consider the case $x, y \neq 0$ and suppose wlog that $xy^{-1} \in A$

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$$v(x+y) = v((xy^{-1}+1)y) = v(xy^{-1}+1) + v(y) \geq v(y) \geq \min(v(x), v(y))$$

The case that one of $x, y = 0$ is trivial. Observe $A = v_A^{-1}(\Gamma_{\geq 0}) \cup \{0\}$. We may therefore construct a one-sided inverse; given a valuation $v : K' \to \Gamma$ we may define $A_v := v^{-1}(\Gamma_{\geq 0}) \cup \{0\}$. It's evident from the two properties of v that A is an additive and multiplicative subgroup. Further by total ordering we see that A is a valuation ring. We only need to demonstrate that the given constructions are mutually inverse, namely $v \sim v_{A_v}$. However this is immediate because the composite $v(K^\star) \to K^\star/A_v^\star \to v_{A_v}(K^\star)$ is an abelian group isomorphism, which preserves the positive segments, and therefore is an order isomorphism. ∎

Corollary 3.25.18 (Valuation Dominance gives a Ring Multiple)
Let $v : K^\star \to \Gamma$ be a valuation with valuation ring A (3.25.17), extended to K by $v(0) := \infty$. If $x, y \in K$ with $y \neq 0$ and $v(x) \geq v(y)$, then $x = ay$ for some $a \in A$.

If further $v(x) = v(y)$ then $a \in A^\star$.

Proof. If $x = 0$ take $a = 0$. Otherwise $x, y \in K^\star$ and $v(xy^{-1}) = v(x) - v(y) \geq 0$, so $a := xy^{-1} \in A$ by (3.25.17), and $x = ay$. If further $v(x) = v(y)$ then $v(a) = 0$, so $a \in A^\star$ by (3.25.17). ∎

Proposition 3.25.19 (Bezout Domain)
Let A be an integral domain. Then the following are equivalent

a) Every finitely generated ideal is principal
b) Every ideal generated by two elements is principal

*In this case we say A is a **Bezout Domain**. Further in this case every pair of elements has a greatest common divisor d which satisfies $d = ax + by$ for some $x, y \in A$.*

Proof. By assumption $(a, b) = (d)$. Immediately we have $d \mid a$ and $d \mid b$. Similarly $d = \lambda a + \mu b$. Suppose $a = xe$ and $b = ye$ then $d = (\lambda x + \mu y)e$ whence $e \mid d$. ∎

Lemma 3.25.20 (Valuation Ring = Local Bezout Domain)
Let A be an integral domain. Then the following are equivalent

a) A is a valuation ring
b) A is a local Bezout domain.

In particular A is a Noetherian valuation ring iff it is a local principal ideal domain.

Proof. Suppose A is a valuation ring and $\mathfrak{a} = (x, y)$. Without loss of generality $xy^{-1} \in A$ whence $x \in (y)$ so $\mathfrak{a} = (y)$ as required.

Conversely suppose A is a local Bezout domain and let $x = \frac{y}{z} \in K$. We may divide by the greatest common divisor so that $\lambda y + \mu z = 1$. We claim one of λy and μz is a unit, for suppose not then by (3.20.3) we have $\lambda y, \mu z \in \mathfrak{m} \implies 1 \in \mathfrak{m}$ a contradiction. Therefore one of y, z is a unit which implies $x \in A$ or $x^{-1} \in A$, and A is a valuation ring. ∎

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